

Giacomo Rossettini: “Placebo, nocebo and contextual effects: the missed link in clinical practice“

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welcome to the fourth active inference Symposium Dr gomo rostini we're very pleased to have you here and um looking forward to your session titled PL ebo nosbo and contextual effects the missing the missed Link in clinical practice over to you thank you for the presentation and for the invitation to use this talk it's a pleasure for me to be here for the hativ inference Symposium and today I talk about clbo noo and contextual effects showing the Mis Link in clinical practice as I mentioned I'm an Italian physical therapist I'm involved in clinici in research and teaching session I come from Verona that is a city in the north of Italy and it represent the city of love on Romeo and Juliet and if you have never visited my suggestion is to do it before to start I want to declare that I have no conflict of interest for my speech I Have organized my presentation to discuss contextual effects firstly we will analyze the definition and the constituent dimensions of contextual effects the mechanisms of action the impact on outcomes and treatments next we will discuss the possible link of contextual effects with the active inference and the implication for clinical practice and research finally we will conclude with a take on message contextual effects are defined as the effects of the context around the therapy and are generated by contextual factors contextual factors were recently defined in international consensus in which I participated in the slide you will find an accept of the final share definition where contextual effector are the perceived views that affect both the patient and the practitioner and can arise from previous experiences and the immediate Dynamics within the therapeutic encounter or a combination of both and from a clinical perspective contextual factors fall into five broad categories or Dimensions that you can see represented in this graph on one hand we have the characteristic of the clinicians with their professional reputations the appearance the beliefs and behaviors on the other side there are the patients characteristic with their expectations preferences and previous experiences between the clinicians and the patients there is the relational Dynamics in terms of verbal and nonverbal

communication what we say and how we communicate in addition to the relational component the treatment characteristics also play an important role they represent the rituality of the treatment finally The Fifth Dimension is represented by the characteristics of the setting within which the encounter is being delivered such as the environment the architecture and also the interior design well all these five Dimensions interacting with each other influence the overall therapeutic outcomes the question we have to ask ourselves is through which mechanisms does this happen I will tell you in the next slides contextual Factors have been studied for their ability to induce Placebo and nocebo effects which represent the effects of the psychosocial context around the treatment and to explain how this occurs I want to start with a clinical example this illustration shows a common clinical encounter in which patients with the muscular skeletal pain receive a therapy such as joint immobilization therapeutic exercises or an ultrasound injections and during the administration of therapy patients are influenced by the specificity of the treatment but also by various different psychosocial cues such as the clinicians work the Sight and Sound of the health care devices the smells of the hygienic and also the taste of the drugs when receiving for example a painkillers well all these cues can influence the patient's perception shaping the experience in a positive or in a negative way triggering Placebo or nocebo effects thus the positive use of contextual factors can produce Placebo effects when these elements are used negatively it produces nocebo effects and the clinical effects of contextual factors can be explained through conscious and unconscious mechanisms of action such as expectation learning processes personality traits mindset and even if genetics however among the most important psychological theories expectation classical conditioning and social learning are the most widely accepted the expectation is a conscious phenomenon through which the patient expects that something will happen in the future we need to consider patients for example who are waiting to receive a therapeutic exercise or manual therapy for a muscular skeletal pain well they will predict a positive therapeutic outcome by evaluating all the contextual factors around the therapy we offer as a clinician such as our Worth or our behaviors instead the classical Pavlovian condition is an unconscious phenomenon concerning the past events in fact patients can learn over time to associate a certain therapy such as for example an injection with a specific context effects that represent the rituality of the technique created by the needles that produce a certain outcome which can be positive or negative but patients can also learn from others they can learn from positive or negative experiences of other patients family members friends or from information they get from social media or blogs regarding for example muscular skeletal pain but how the contextual Factor

Works in practice I will use this image to explain this mechanisms of action briefly whenever patient interact with the healthcare context that are always influenced by the five contextual factors and these elements are encoded by the patient brain who interprets them and provides an evaluation and a prediction of their meaning and if this interpretation is positive Place effects are triggered if the interpretation is negative noo effects are stimulated and these effects are sustained by a Cascade of a psychon immuno endocrinological events at multiple level of the central nervous systems in terms of the release neurotransmitters for example negative contextual factors can activate the release of kyin and the deactivation of dopamine and the opioid systems instead the use of the positive Context factors can facilitate the release of a PES endocanids andam and also different brain areas can be activated for example the dorsal lateral peral cortex the anterior singulate cortex and the g per gray and the dorsal Horn of the spine are modulated and all these specific neurological Pathways translated into changes in clinical outcomes and using for example muscul scal pain as an example if the context oper is positive the patients will perceive a relief of symptoms instead if the Context adopted is negative the patient will have an aggravation of symptoms and contextual factors interacting with the treatment specificity are able to influence more the patient subjective outcomes or the illness demensions than the objective outcomes that represent the disease component for example pain disability experience satisfaction represent typical example of outcomes influenced by contextual factors however not all the changes of the therapeutic outs are due by contextual factors or by the specificity of our treatment in fact several Compounders can explain the improvements or the worsening of the patient clinical condition an example are reported in this graph for example the natural history of the disease can explain the natural flation of the of symtom or the regression towards the means which represent a statistical phenomena or for example the presence of unidentified intervention as in the case of patient that present a muscular schal pain or receive a manual therapy or therapeutic exercise and concomitantly receive also a painkillers another question we should have ask ourself is what is the weight of contextual factors influences the outcome and for a long time it was a common opinion in the scientific Comm that about 33 to 4 35% of the total therapeutic effects depend on the contextual component and even if it me seem limited it is actually not because it is a component influencing clinical outcome that we should know and we should use in our clinical practice and the international debate on the weight of contextual effects has continued to the present day without find the conclusion several recent systematic reviews with the with the metanalysis has assessed the weight of contextual effects what the different authors have

observed is that the contextual effects can vary from small to large and I report here different example you can see what happen for no specific loic pain neck pain fibromyalgia and also NEOS arthritis and the heterogeneity in these results can be explained for example by the characteristic of the patients the clinicians by the characteristic of the condition analyze the interventions considered and also by the characteristic of the statistical metric adopted so some student has also provided data on the contextual effects for treatments for example in this meta research the contextual effects of of different treatments was evaluated and what we can observe is that the contextual is progressively higher in treatment with the high realistic Futures to go into details in psychological treatments the the contextual effects account for the 38% in pharmacological treatments it accounts for the 61% and in the physical treatment such as manual therapy or handson techniques it accounts for the 65% 64% and having reached this point I think that we need to understand how contextual effects can be related to the theme of this coner this conference the active inference and to understand this we need to start with a clinical example we need to think about the management of patients with persistent muscular skial pain who have negative expectations based on their previous experiences psychological characteristics or personality traits and for example if you consider a patient with the neck pain that in the past has been exposed to exercises and learn that these exercises produces pain consolidating a negative internal model to worth this in our patients Clinic we offer exercises to our neck pain patients in accordance with the best evidence and as a result of this direct experience the patient can have two main outcomes at first for example it can confirm that exercises can produce pain or the patients may not perceive the pain violated their expectations and this violation of expectation is interesting because it can result in two main scenarios one adapted IND where patients Mak an it makes an update of their expectation and on their internal model and a d functional one where the patients consider this violation as an exception to the rule or as results to the chance and therefore maintain it and this is referred to as a cognitively immuniz patients and we suggest that such differen in updating responses can be understood it at least to some extent from a basian perspectiv and in fact the ability to make an update of the patient expectation can be interpreted in the light of the basian model and will depend on the strength and accuracy of the patient prior beliefs and behaviors and from a biiian perspectives our brain is conceptualizes a predictor machine that generates predictions know as PRI about the expected sensory inputs and the integration between the prior and the sensor inputs results in the posterior the percept with which can be more or less influenced by the prior and by the sensory data depending on their level of precision and let let us think

about the patients with the pain who has negative prior or expectations toward exercise they represented the sensory input and if these priors are very precise and reliable it will be much more difficult for the patient to rely on the sensory input given by the exercises and the prediction error is really minimal and the patients consider the experience as a result of a of the chance or an exception of the rule existing in maintenance or the initial expectations instead if the interpretation are less precise and reliable it will be much easier for the patients to rely on the sensor inputs given by the exercise because the prediction error between what it is expected and what it is experienced will be more significant will contribute to improving the patient expectation by generating update and I think is very fascinating and complex but it does all us to think about how we can use the context to challenge or Make an updates of the patients prior and I will try to help you to understand this point with another clinical example with this B model in mind it is possible to better understand the clinical scenario presented in this image where we have the patients with the muscular pain of the shoulder has had a negative PR in that in that situation the patients prefer not to rise their harms because they have been told that the tendons can stop on the other hand we have the clinicians who tries to challenge the patients prior through the tools that he has disposal for example the exercises the manual therapy the virtual reality and the education but the clinician also has another secret weapon to use that this school has the contextual factors made by words cues behaviors rituality that can influence the patient's brain and the patients prior and as an outcome of this challenge we can have for example expectation violation following by an update which is realized when the prior with low Precision of our patient is updated accordingly with the newly acquired evidence the patients leave the bre the arms without any pain and without any types of lesions another possible scenario is represented by the expectation violation followed by the maintenance of the dysfunctional beliefs as the consequence of one having a highly precise prior which is considered really reliable and therefore the newly acquired information and this confirmatory evidence is not sufficient to disprove and update the such strong prior and for example in this case we have a patient with a muscular pain that during a clinical session manage to leave their arm without any negative consequences to their shoulder but still believe that leaving the arm will possibly create and lead to the shoulder to break are likely to have for example a very high precise prior and might discard the positive evidence for example the success in lifting the arm which might be classified as an exception instead of the rule so try to pull the string of the speech what might be the implication for this uh from this talk to our clinical practice the use of contextual factors represent an opportunity to Assess the evidence

based therapy we know it as a clinician but as a clinician we also need to consider that using context of factors such as words touch the rituality of the technique and environment could help to challenge the patients prior and improve the patient muscular Cal pain however this observation that arise from the clinic need to be supported by data and more study to substantiate their effectiveness thus researchers are called up to integrate contextual effects into the study on muscal pain using for example the basian brain and the active inference lens and to do is this it is important to produce experimental work in the lab in the click work in patient to validate these theoretical models by using for example subjective outcomes but also some objective physiological metrics we are in the end so I concluded I want to cre in you wishing you to remain curious clinicians capable to see the things from different perspective and asset the complexity presented in the clinic and also within this image let's not just focus only on the details the te the technique it is important but it is not all let us always remember that the scenario in which we operate it is much more complex and both the context the moon and the confounders The Surge and the sky comes into play only if we Embrace this opportunity and this complexity we could find new strategy to best manage our patients with muscular skial pain thanks for your attentions thank you thank you till next time